

EDEXCEL INTERNATIONAL A LEVEL

# WMA11/01 October 2024 Worked Solutions

Oct 2024 paper rebuilt with the worked solution after each question.

**9**

paper questions

**WMA11**

Pure 1

**Single-  
paper  
solutions**standalone IAL  
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - [eliteigcse.com](https://eliteigcse.com)

**P1**

**PAST PAPER**

# WMA11/01 October 2024

Oct 2024 | 9 questions | 75 marks

**9**

questions

**75**

marks

**Answers**

worked solution  
after each  
question

## Question 1

Straight Line

1. The line  $l_1$  passes through the point  $A(-5, 20)$  and the point  $B(3, -4)$ .

- (a) Find an equation for  $l_1$  giving your answer in the form  $y = mx + c$ , where  $m$  and  $c$  are constants.

(3)

The line  $l_2$  is perpendicular to  $l_1$  and passes through the midpoint of  $AB$

- (b) Find an equation for  $l_2$  giving your answer in the form  $px + qy + r = 0$ , where  $p$ ,  $q$  and  $r$  are integers.

(3)

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(Total for Question 1 is 6 marks)

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## Worked Solution - Question 1

**1. Find the gradient of  $l_1$** 

$$m = \frac{-4 - 20}{3 - (-5)} = \frac{-24}{8} = -3.$$

**2. Find the equation of  $l_1$** 

Using  $B(3, -4)$ ,  $y = -3x + c$ . Then  $-4 = -9 + c$ , so  $c = 5$ .

**3. Find the midpoint of  $AB$** 

$$M = \left( \frac{-5 + 3}{2}, \frac{20 + (-4)}{2} \right) = (-1, 8).$$

**4. Use perpendicular gradient**

Since  $l_1$  has gradient  $-3$ ,  $l_2$  has gradient  $\frac{1}{3}$ .

**5. Find the equation of  $l_2$** 

$$y - 8 = \frac{1}{3}(x + 1), \text{ so } 3y - 24 = x + 1 \text{ and } x - 3y + 25 = 0.$$

**Final answer**

(a)  $y = -3x + 5$ .

(b)  $x - 3y + 25 = 0$ .

## Question 2

## Indices &amp; Surds

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) Simplify fully

$$\frac{3y^3(2x^4)^3}{4x^2y^4} \quad (3)$$

(ii) Find the exact value of  $a$  such that

$$\frac{16}{\sqrt{3} + 1} = a\sqrt{27} + 4$$

Write your answer in the form  $p\sqrt{3} + q$  where  $p$  and  $q$  are fully simplified rational constants.

(4)

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(Total for Question 2 is 7 marks)

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## Worked Solution - Question 2

**1. Simplify the power bracket**

$$(2x^4)^3 = 8x^{12}.$$

**2. Simplify the algebraic fraction**

$$\frac{3y^3(2x^4)^3}{4x^2y^4} = \frac{3y^3(8x^{12})}{4x^2y^4} = \frac{6x^{10}}{y}.$$

**3. Rationalise the left side**

$$\frac{16}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = 8\sqrt{3} - 8.$$

**4. Equate the surd parts**

$$8\sqrt{3} - 8 = a\sqrt{27} + 4 = 3a\sqrt{3} + 4.$$

**5. Solve for a**

$$3a\sqrt{3} = 8\sqrt{3} - 12, \text{ so } a = \frac{8\sqrt{3} - 12}{3\sqrt{3}} = \frac{8}{3} - \frac{4}{3}\sqrt{3}.$$

**Final answer**

$$(i) \frac{6x^{10}}{y}. \quad (ii) a = -\frac{4}{3}\sqrt{3} + \frac{8}{3}.$$

## Question 3

## Integration

3.

In this question you must show all stages of your working.

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} \quad x > 0$$

(a) Find  $\int f(x) \, dx$ 

(4)

(b) (i) Show that when  $f'(x) = 0$ 

$$3x^2 + 10x - 25 = 0$$

(ii) Hence state the value of  $x$  for which

$$f'(x) = 0$$

(5)

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(Total for Question 3 is 9 marks)

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## Worked Solution - Question 3

**1. Rewrite  $f(x)$** 

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} = \frac{x^2 + 10x + 25}{x^{1/2}} = x^{3/2} + 10x^{1/2} + 25x^{-1/2}.$$

**2. Integrate**

$$\int f(x) dx = \frac{2}{5}x^{5/2} + \frac{20}{3}x^{3/2} + 50x^{1/2} + c.$$

**3. Differentiate  $f(x)$** 

$$f'(x) = \frac{3}{2}x^{1/2} + 5x^{-1/2} - \frac{25}{2}x^{-3/2}.$$

**4. Set  $f'$  prime equal to zero**

$$\frac{3}{2}x^{1/2} + 5x^{-1/2} - \frac{25}{2}x^{-3/2} = 0.$$

**5. Clear powers of  $x$** 

Multiplying by  $2x^{3/2}$  gives  $3x^2 + 10x - 25 = 0$ .

**6. Use  $x$  greater than zero**

$$3x^2 + 10x - 25 = (3x - 5)(x + 5), \text{ so } x = \frac{5}{3} \text{ or } x = -5. \text{ Since } x > 0, x = \frac{5}{3}.$$

**Final answer**

$$(a) \frac{2}{5}x^{5/2} + \frac{20}{3}x^{3/2} + 50x^{1/2} + c.$$

$$(b) x = \frac{5}{3}.$$

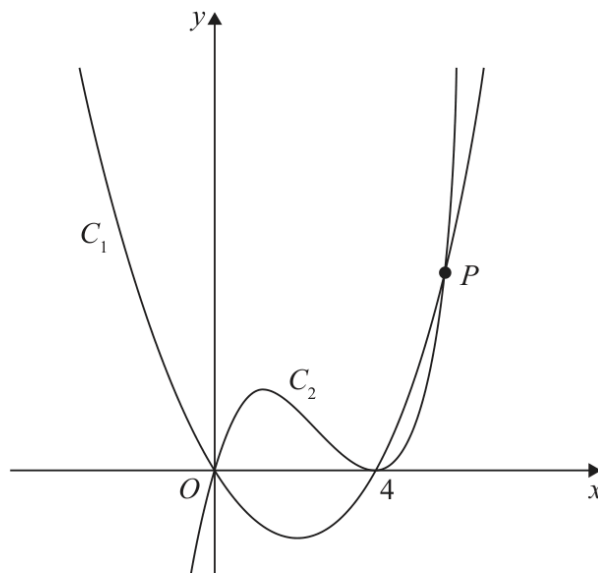
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10 marks

## Question 4

Polynomials

4.



**Figure 1**

Figure 1 shows a sketch of part of the curves  $C_1$  and  $C_2$

Given that  $C_1$

- has equation  $y = f(x)$  where  $f(x)$  is a quadratic function
- cuts the  $x$ -axis at the origin and at  $x = 4$
- has a minimum turning point at  $(2, -4.8)$

(a) find  $f(x)$

**(3)**

Given that  $C_2$

- has equation  $y = g(x)$  where  $g(x)$  is a cubic function
- cuts the  $x$ -axis at the origin and meets the  $x$ -axis at  $x = 4$
- passes through the point  $(6, 7.2)$

(b) find  $g(x)$

**(3)**

The curves  $C_1$  and  $C_2$  meet in the first quadrant at the point  $P$ , shown in Figure 1.

(c) Use algebra to find the coordinates of  $P$ .

**(4)**

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**(Total for Question 4 is 10 marks)**

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## Worked Solution - Question 4

**1. Use the roots of  $C_1$** 

Since  $C_1$  cuts the x-axis at 0 and 4, let  $f(x) = ax(x - 4)$ .

**2. Use the minimum point of  $C_1$** 

At  $x = 2$ ,  $f(2) = -4a = -4.8 = -\frac{24}{5}$ , so  $a = \frac{6}{5}$ .

**3. Write  $f(x)$** 

$$f(x) = \frac{6}{5}x(x - 4).$$

**4. Use the roots of  $C_2$** 

The cubic cuts the x-axis at the origin and meets the x-axis at  $x = 4$ , so  $x = 4$  is a repeated root. Let  $g(x) = bx(x - 4)^2$ .

**5. Use the point (6, 7.2)**

$$7.2 = b(6)(2^2) = 24b, \text{ so } b = 0.3 = \frac{3}{10}.$$

**6. Equate the curves**

$$\frac{6}{5}x(x - 4) = \frac{3}{10}x(x - 4)^2.$$

**7. Find the first-quadrant intersection**

Ignoring the shared roots on the x-axis, divide by  $x(x - 4)$  to get

$$\frac{6}{5} = \frac{3}{10}(x - 4). \text{ Hence } x = 8.$$

**8. Find  $y$** 

$$y = f(8) = \frac{6}{5}(8)(4) = \frac{192}{5}.$$

**Final answer**

$$(a) f(x) = \frac{6}{5}x(x - 4).$$

$$(b) g(x) = \frac{3}{10}x(x - 4)^2.$$

$$(c) P = (8, \frac{192}{5}).$$

## Question 5

Radians

5. A plot of land  $OAB$  is in the shape of a sector of a circle with centre  $O$ .

Given

- $OA = OB = 5 \text{ km}$
- angle  $AOB = 1.2 \text{ radians}$

- (a) find the perimeter of the plot of land.

(2)

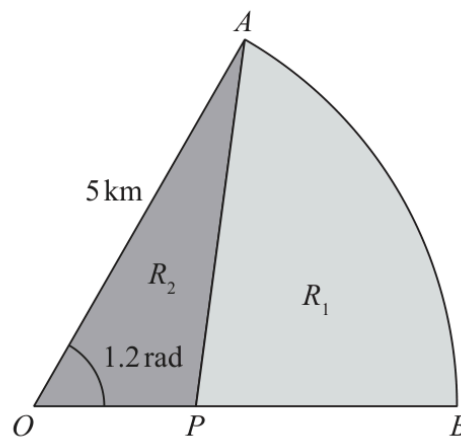


Diagram **NO**  
accurately dr

**Figure 2**

A point  $P$  lies on  $OB$  such that the line  $AP$  divides the plot of land into two regions  $R_1$  and  $R_2$  as shown in Figure 2.

Given that

$$\text{area of } R_1 = 3 \times \text{area of } R_2$$

- (b) show that the area of  $R_2 = 3.75 \text{ km}^2$

(3)

- (c) Find the length of  $AP$ , giving your answer to the nearest 100 m.

(4)

(Total for Question 5 is 9 marks)

## Worked Solution - Question 5

**1. Find the arc length**

$$\text{arc } AB = r\theta = 5(1.2) = 6 \text{ km.}$$

**2. Find the perimeter**

$$OA + OB + \text{arc } AB = 5 + 5 + 6 = 16 \text{ km.}$$

**3. Find the total sector area**

$$\text{Area of } OAB = \frac{1}{2}(5^2)(1.2) = 15 \text{ km}^2.$$

**4. Use the area ratio**

Since  $\text{area } R_1 = 3 \times \text{area } R_2$ , the total area is  $4 \times \text{area } R_2$ .

**5. Find area R2**

$$\text{area } R_2 = \frac{15}{4} = 3.75 \text{ km}^2.$$

**6. Find OP from triangle OAP**

$$\frac{1}{2}(5)(OP) \sin 1.2 = 3.75, \text{ so } OP \approx 1.609 \text{ km.}$$

**7. Use the cosine rule**

$$AP^2 = 5^2 + OP^2 - 2(5)(OP) \cos 1.2, \text{ so } AP \approx 4.665 \text{ km.}$$

**8. Round to the nearest 100 m**

$$AP = 4.7 \text{ km to the nearest } 0.1 \text{ km.}$$

**Final answer**

(a) 16 km.

(b)  $3.75 \text{ km}^2$ .

(c)  $AP = 4.7 \text{ km}$ .

## Question 6

## Graphs of Functions

6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve  $C$  with equation

$$y = \frac{1}{2-x} \quad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of  $C$  with the  $y$ -axis

(3)

The straight line  $l$  has equation  $y = kx - 4$ , where  $k$  is a constant.

Given that  $l$  cuts  $C$  at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geq 0$$

(ii) find the range of possible values for  $k$ .

(6)

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(Total for Question 6 is 9 marks)

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## Worked Solution - Question 6

Topic group

**1. Identify the graph features**

$y = \frac{1}{2-x}$  has vertical asymptote  $x = 2$  and horizontal asymptote  $y = 0$ .

**2. Find the y-intercept**

At  $x = 0$ ,  $y = \frac{1}{2}$ , so the y-intercept is  $(0, \frac{1}{2})$ .

**3. Set up the intersection with the line**

If  $y = kx - 4$  cuts  $C$ , then  $kx - 4 = \frac{1}{2-x}$ .

**4. Form the quadratic**

$(kx - 4)(2 - x) = 1$ , so  $kx^2 - (2k + 4)x + 9 = 0$ .

**5. Use the discriminant**

For at least one intersection, the discriminant is non-negative.

**6. Derive the given inequality**

$(2k + 4)^2 - 4(k)(9) \geq 0$ , so  $4(k^2 - 5k + 4) \geq 0$ , hence  $k^2 - 5k + 4 \geq 0$ .

**7. Solve the inequality**

$k^2 - 5k + 4 = (k - 1)(k - 4)$ , so  $k \leq 1$  or  $k \geq 4$ .

**Final answer**

(a) vertical asymptote  $x = 2$ , y-intercept  $(0, \frac{1}{2})$ .

(b)  $k \leq 1$  or  $k \geq 4$ .

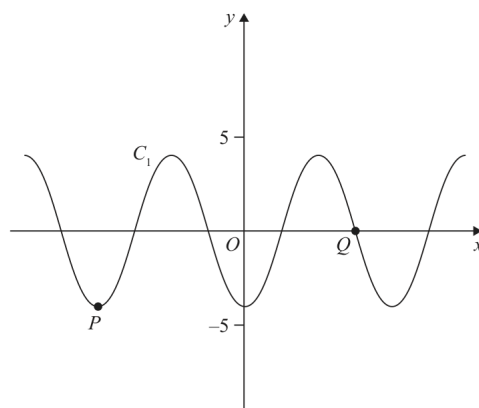
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8 marks

## Question 7

Trigonometric Functions

7.



**Figure 3**

Figure 3 shows a plot of part of the curve  $C_1$  with equation

$$y = -4 \cos x$$

where  $x$  is measured in radians.

Points  $P$  and  $Q$  lie on the curve and are shown in Figure 3.

(a) State

- (i) the coordinates of  $P$
- (ii) the coordinates of  $Q$

(3)

The curve  $C_2$  has equation  $y = -4 \cos x + k$  where  $x$  is measured in radians and  $k$  is a constant.

Given that  $C_2$  has a maximum  $y$  value of 11

(b) (i) state the value of  $k$

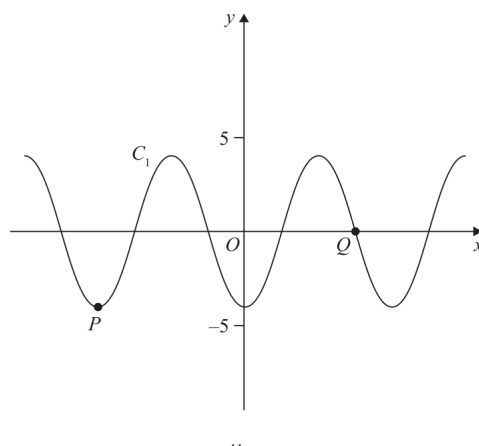
- (ii) state the coordinates of the minimum point on  $C_2$  with the smallest positive  $x$  coordinate.

(3)

On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4 \cos x = 5 - \frac{10}{\pi} x$$



(Total for Question 7 is 8 marks)

# Worked Solution - Question 7

## 1. Use $y$ equals negative $4 \cos x$

For  $y = -4 \cos x$ , minimum points occur at  $x = 2\pi n$  with value  $-4$ .

## 2. Read P

The displayed minimum to the left of the  $y$ -axis is  $P = (-2\pi, -4)$ .

## 3. Read Q

The point  $Q$  is the positive  $x$ -intercept after the maximum at  $x = \pi$ , so  
 $Q = (\frac{3\pi}{2}, 0)$ .

## 4. Find $k$ for C2

For  $y = -4 \cos x + k$ , the maximum value is  $4 + k$ . Given the maximum is  $11$ ,  
 $k = 7$ .

## 5. Find the smallest positive minimum on C2

The minimum value is  $-4 + 7 = 3$ . The first positive minimum is at  $x = 2\pi$ , so  
the point is  $(2\pi, 3)$ .

## 6. Interpret the equation graphically

$-4 \cos x = 5 - \frac{10}{\pi}x$  is the intersection of the cosine curve with the straight line  
 $y = 5 - \frac{10}{\pi}x$ .

## 7. Count intersections

The line passes through  $(0, 5)$  and meets the displayed curve once, at the  $x$ -axis point, so there is 1 solution.

**Final answer**

$(a)(i) P = (-2\pi, -4).$      $(a)(ii) Q = (\frac{3\pi}{2}, 0).$

$(b)(i) k = 7.$

$(b)(ii) (2\pi, 3).$

$(c)$  1 solution.

## Question 8

## Integration

8. A curve  $C$  has equation  $y = f(x)$ .

The point  $P$  with  $x$  coordinate 3 lies on  $C$

Given

- $f'(x) = 4x^2 + kx + 3$  where  $k$  is a constant
- the normal to  $C$  at  $P$  has equation  $y = -\frac{1}{24}x + 5$

(a) show that  $k = -5$

(3)

(b) Hence find  $f(x)$ .

(4)

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(Total for Question 8 is 7 marks)

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## Worked Solution - Question 8

**1. Find the y-coordinate of P**

The normal is  $y = -\frac{1}{24}x + 5$ . At  $x = 3$ ,  $y = -\frac{3}{24} + 5 = \frac{39}{8}$ .

**2. Use the normal gradient**

The normal gradient is  $-\frac{1}{24}$ , so the tangent gradient is **24**.

**3. Use f prime at x equals 3**

$$f'(3) = 4(3)^2 + 3k + 3 = 39 + 3k.$$

**4. Find k**

$$39 + 3k = 24, \text{ so } k = -5.$$

**5. Integrate f prime**

With  $k = -5$ ,  $f'(x) = 4x^2 - 5x + 3$ , so  $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x + c$ .

**6. Use point P**

$$\frac{39}{8} = \frac{4}{3}(27) - \frac{5}{2}(9) + 3(3) + c = \frac{45}{2} + c.$$

**7. Find c**

$$c = \frac{39}{8} - \frac{45}{2} = -\frac{141}{8}.$$

**Final answer**

$$k = -5 \text{ and } f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x - \frac{141}{8}.$$

## Question 9

## Differentiation

9.

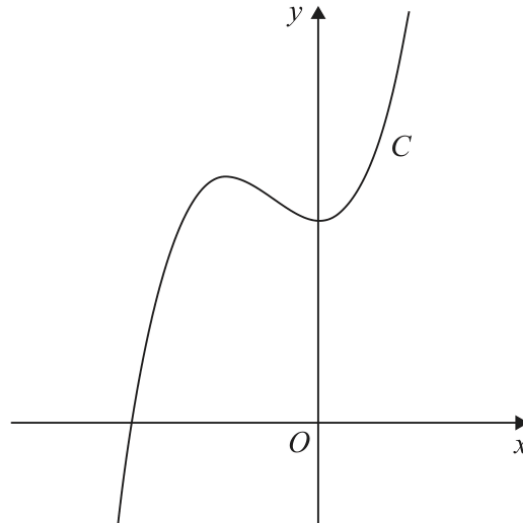


Figure 4

Figure 4 shows a sketch of the curve  $C$  with equation  $y = f(x)$ , where

$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

- (a) Deduce the range of values of  $x$  for which  $f(x) \geq 0$  (1)
- (b) Find  $f'(x)$  giving your answer in simplest form. (3)

The point  $R(-4, 84)$  lies on  $C$ .

Given that the tangent to  $C$  at the point  $P$  is parallel to the tangent to  $C$  at the point  $R$

- (c) find the  $x$  coordinate of  $P$ . (4)
- (d) Find the point to which  $R$  is transformed when the curve with equation  $y = f(x)$  is transformed to the curve with equation,
- (i)  $y = f(x - 3)$
- (ii)  $y = 4f(x)$  (2)

(Total for Question 9 is 10 marks)

## Worked Solution - Question 9

**1. Use the positive quadratic factor**

$3x^2 - 4x + 20$  has discriminant  $(-4)^2 - 4(3)(20) < 0$  and positive leading coefficient, so it is always positive.

**2. Deduce the sign of  $f(x)$** 

The sign of  $f(x) = (x + 5)(3x^2 - 4x + 20)$  is therefore the sign of  $x + 5$ .  
Hence  $f(x) \geq 0$  for  $x \geq -5$ .

**3. Expand  $f(x)$** 

$$f(x) = 3x^3 + 11x^2 + 100.$$

**4. Differentiate**

$$f'(x) = 9x^2 + 22x.$$

**5. Find the gradient at  $R$** 

$$\text{At } R(-4, 84), f'(-4) = 9(16) + 22(-4) = 56.$$

**6. Use parallel tangents**

For the tangent at  $P$  to be parallel,  $f'(x) = 56$ .

**7. Solve for  $x$** 

$$9x^2 + 22x - 56 = 0, \text{ so } x = \frac{-22 \pm 50}{18}. \text{ The roots are } x = -4 \text{ and } x = \frac{14}{9}.$$

**8. Choose the point  $P$** 

$$x = -4 \text{ is the point } R, \text{ so } x_P = \frac{14}{9}.$$

### 9. Transform R under y equals f of x minus 3

$y = f(x - 3)$  translates the curve 3 units to the right, so  $(-4, 84)$  maps to  $(-1, 84)$ .

### 10. Transform R under y equals 4f(x)

$y = 4f(x)$  stretches vertically by scale factor 4, so  $(-4, 84)$  maps to  $(-4, 336)$ .

#### Final answer

(a)  $x \geq -5$ .

(b)  $f'(x) = 9x^2 + 22x$ .

(c)  $x_P = \frac{14}{9}$ .

(d)(i)  $(-1, 84)$ .

(d)(ii)  $(-4, 336)$ .